

In this lecture block we will study how predictive models support the development of sustainable chemical systems. Together we will:

- dive into the foundations of predictive modelling
- develop data-driven models for predicting molecular properties
- apply the models for selecting solvents in line with green chemistry principles

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- background in **process engineering**
 - Bachelor in **Biosystems Engineering** (Otto von Guericke University Magdeburg)
 - Master in **Systems Engineering and Engineering Cybernetics** (Otto von Guericke University Magdeburg)
- PhD and research on **computer-aided molecular and process design for biorefineries**

Course materials

All course materials including slides, data, and code can be found on GitHub: **[Sustainable Chemical Systems Course](#)**
Download the course contents. Navigate to "html_slides" and open them in your browser.

Installation

All installation instructions can be found in the README.md. Installation will be addressed later in the course as well.

Literature

Online Courses:

- Deep Learning Specialization by Andrew Ng, K. Katanforoosh and B. Mourri from Coursera learning platform: A course series that requires a minimum of mathematical skills but still brings you from zero to a very good understanding and practical training on deep learning.

Books:

- Deep Learning by I. Goodfellow, Y. Bengio and A. Courville: Open access, authored by pioneers on deep learning.
- Hands-On Machine Learning with Scikit-Learn and TensorFlow by A. Geron: For people who prefer learning by doing.

What is a model?

There are different definitions for "model" across disciplines

Philosophy

According to Herbert Stachowiak, a model is characterized by at least three properties:

1. Mapping A model always is a model of something—it is an image or representation of some natural or artificial, existing or imagined original,[11] where this original itself could be a model.
2. Reduction In general, a model will not include all attributes that describe the original but only those that appear relevant to the model's creator or user.
3. Pragmatism A model does not relate unambiguously to its original. It is intended to work as a replacement for the original a) for certain subjects (for whom?) b) within a certain time range (when?) c) restricted to certain conceptual or physical actions (what for?).

Mathematics

A mathematical model is an abstract description of a concrete system using mathematical concepts and language. The process of developing a mathematical model is termed mathematical modeling. Mathematical models are used in many fields, including applied mathematics, natural sciences, social sciences[1][2] and engineering. In particular, the field of operations research studies the use of mathematical modelling and related tools to solve problems in business or military operations. A model may help to characterize a system by studying the effects of different components, which may be used to make predictions about behavior or solve specific problems.

Why do we need models in sustainable chemistry and chemical engineering?

- predict pure compound properties
- predict mixture properties
- molecular design
- process design
- catalyst design
- tune reactions
- toxicity predictions

Predicting chemical properties before synthesis

Boiling point, melting point, solubility, logP, toxicity.

Saves time, money, and reduces hazardous experiments.

Example: Screening bio-based solvents for “green” alternatives without making dozens of compounds in the lab.

Catalyst design

Mechanistic or data-driven models predict activity/selectivity.

Example: Machine learning models predict which metal-ligand combinations will give high yield for a desired reaction.

Reaction optimization

Reaction conditions (temperature, solvent, catalyst load) optimized via models.

Example: Bayesian optimization can find the best conditions in <10 experiments instead of hundreds.

Process design and scale-up

Models predict mass/energy balances, separation efficiency, and environmental impact.

Example: Designing a biorefinery to maximize product yield while minimizing waste.

Circular chemistry / life-cycle assessment

Models predict carbon footprint or energy intensity of processes.

Example: Comparing traditional vs bio-based routes to make the same chemical.

Drug discovery

QSAR/QSPR models predict biological activity, ADMET properties.

Screening large virtual libraries impossible without predictive models.

neue prozesse entwefen und auslegen

neue prozesse verschalten

prozesssysteme optimieren

Example: Chemical space: find the right molecule for the desired application

What makes a good model?

MENTIMETER

different scales different accuracies

atom level molecular level phase level process level plant level

examples which techniques to use (maybe do it as a graphic)

Quote: "All models are wrong, but some are useful"

Good model: predictive, generalizable, interpretable, fit-for-purpose

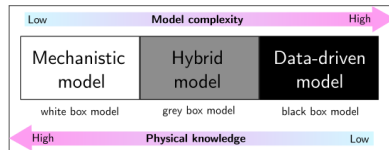
Modelling scales

graphic scale, modelling approach

Modelling approaches

Cow example

Mechanistic vs hybrid vs data-driven model



show on cow example

show on real example



from [wikipedia](#) (Oeschinensee_Slaunger_2009-07-07.jpg), public domain

Linear vs. nonlinear

If all the operators in a mathematical model exhibit linearity, the resulting mathematical model is defined as linear. All other models are considered nonlinear. The definition of linearity and nonlinearity is dependent on context, and linear models may have nonlinear expressions in them. For example, in a statistical linear model, it is assumed that a relationship is linear in the parameters, but it may be nonlinear in the predictor variables. Similarly, a differential equation is said to be linear if it can be written with linear differential operators, but it can still have nonlinear expressions in it. In a mathematical programming model, if the objective functions and constraints are represented entirely by linear equations, then the model is regarded as a linear model. If one or more of the objective functions or constraints are represented with a nonlinear equation, then the model is known as a nonlinear model. Linear structure implies that a problem can be decomposed into simpler parts that can be treated independently or analyzed at a different scale, and therefore that the results will remain valid if the initial is recomposed or rescaled.

Nonlinearity, even in fairly simple systems, is often associated with phenomena such as chaos and irreversibility. Although there are exceptions, nonlinear systems and models tend to be more difficult to study than linear ones. A common approach to nonlinear problems is linearization, but this can be problematic if one is trying to study aspects such as irreversibility, which are strongly tied to nonlinearity. Static vs. dynamic

A dynamic model accounts for time-dependent changes in the state of the system, while a static (or steady-state) model calculates the system in equilibrium, and thus is time-invariant. Dynamic models are typically represented by differential equations or difference equations. Explicit vs. implicit

If all of the input parameters of the overall model are known, and the output parameters can be calculated by a finite series of computations, the model is said to be explicit. But sometimes it is the output parameters which are known, and the corresponding inputs must be solved for by an iterative procedure, such as Newton's method or Broyden's method. In such a case the model is said to be implicit. For example, a jet engine's physical properties such as turbine and nozzle throat areas can be explicitly calculated given a design thermodynamic cycle (air and fuel flow rates, pressures, and temperatures) at a specific flight condition and power setting, but the engine's operating cycles at other flight conditions and power settings cannot be explicitly calculated from the constant physical properties. Discrete vs. continuous

A discrete model treats objects as discrete, such as the particles in a molecular model or the states in a statistical model; while a continuous model represents the objects in a continuous manner, such as the velocity field of fluid in pipe flows, temperatures and stresses in a solid, and electric field that applies continuously over the entire model due to a point charge. Deterministic vs. probabilistic (stochastic)

A deterministic model is one in which every set of variable states is uniquely determined by parameters in the model and by sets of previous states of these variables; therefore, a deterministic model always performs the same way for a given set of initial conditions. Conversely, in a stochastic model—usually called a "statistical model"—randomness is present, and variable states are not described by unique values, but rather by probability distributions. Deductive, inductive, or floating

A deductive model is a logical structure based on a theory. An inductive model arises from empirical findings and generalization from them. If a model rests on neither theory nor observation, it may be described as a 'floating' model. Application of mathematics in social sciences outside of economics has been criticized for unfounded models.[3] Application of catastrophe theory in science has been characterized as a floating model.[4] Strategic vs. non-strategic

Models used in game theory are distinct in the sense that they model agents with incompatible incentives, such as competing species or bidders in an auction. Strategic models assume that players are autonomous decision makers who rationally choose actions that maximize their objective function. A key challenge of using strategic models is defining and computing solution concepts such as the Nash equilibrium. An interesting property of strategic models is that they separate reasoning about rules of the game from reasoning about behavior of the players.[5]

Data-driven modelling

AI vs ML vs DL -> Begriffe erklären

supervised vs unsupervised vs reinforcement learning

architectures: types of models (MLP, DNN, GNN, Transformers)